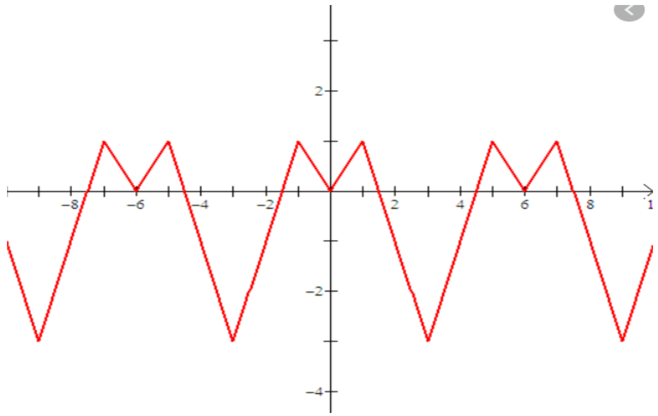
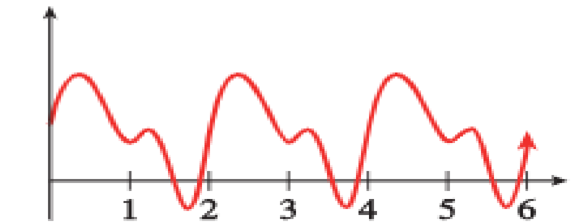
B. Tech – II

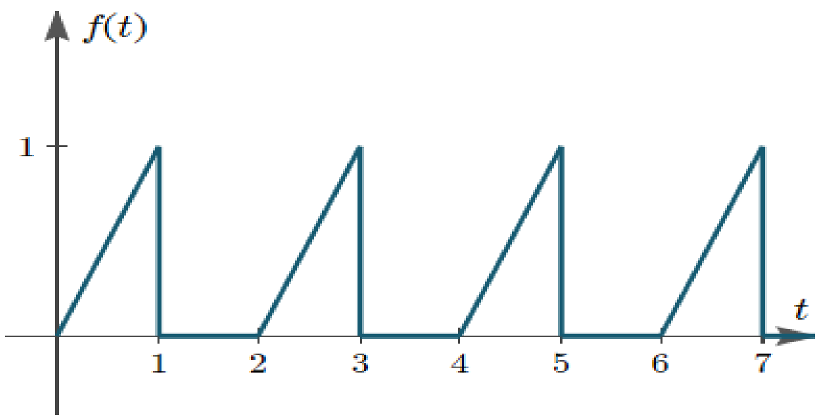
Department of Science and Humanities

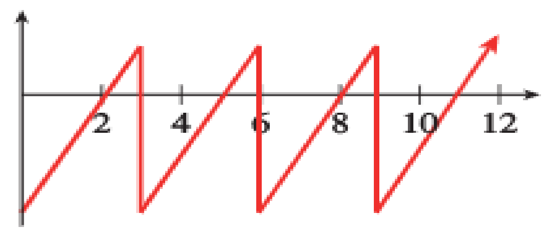
WHAT DO YOU OBSERVE FROM GIVEN GRAPH ?





(a)



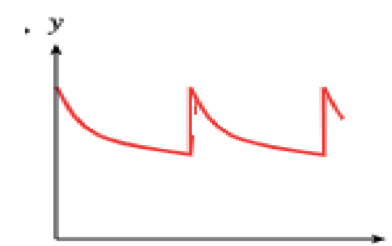


PERIODIC FUNCTION

A ***periodic function*** is one whose values repeat at evenly spaced intervals, or ***periods***, of the input variable. Periodic functions are used to model phenomena that exhibit cyclical behavior, such as growth patterns in plants and animals, radio waves, and planetary motion. we consider some applications of periodic functions.

EXAMPLE

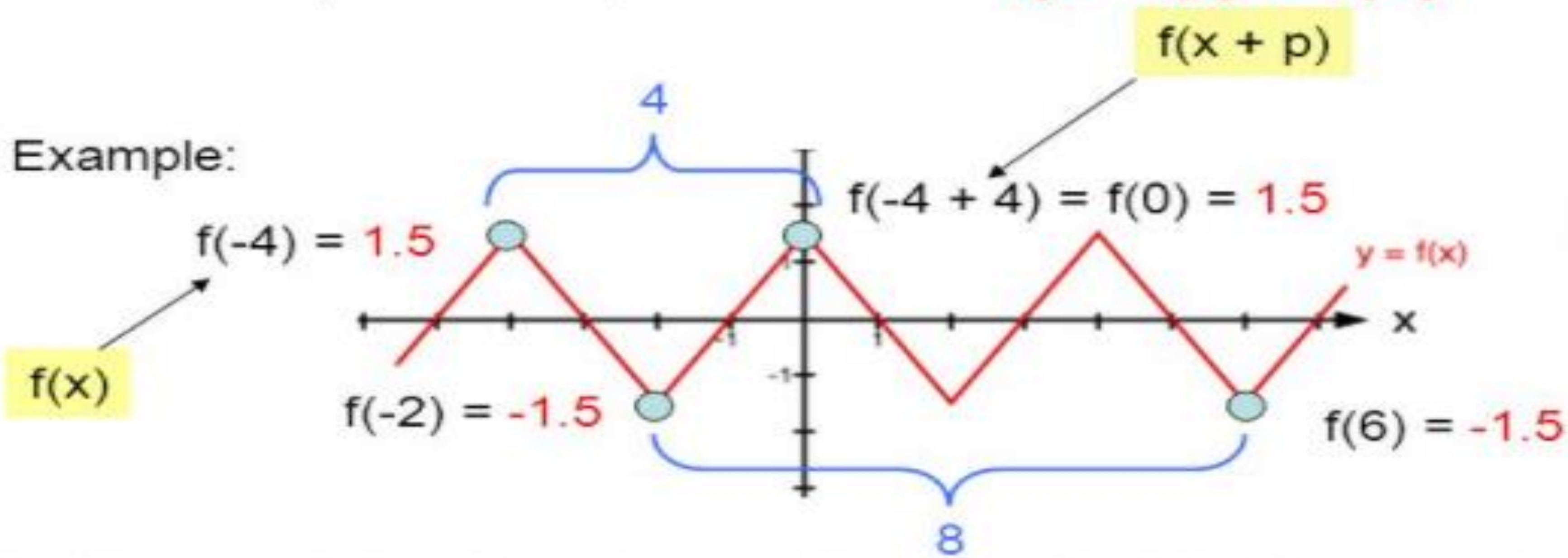
When the heart contracts, blood pressure in the arteries rises rapidly to a peak (systolic blood pressure) and then falls off quickly to a minimum (diastolic blood pressure). Blood pressure is a periodic function of time.



PERIOD OF PERIODIC FUNCTION

- **Definition**: A function f is **periodic** if there is a positive number p, called a period of f, such that $\mathbf{f(x + p) = f(x)}$

Example:



periodic

Definition: The **fundamental period** is the smallest period of a periodic function.

Here the **fundamental period is 4.**

HOW DO WE TAKE LAPLACE TRANSFOR FOR PERIODIC FUNCTION

THEOREM

Let f be periodic, that is:

$$\exists T \in \mathbb{R}_{\neq 0} : \forall x \in \mathbb{R} : f(x) = f(x + T)$$

Then:

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t)\, \mathrm{d}t$$

where $\mathcal{L}\{f(t)\}$ denotes the Laplace transform.

PROOF

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t)\ \mathrm{d}t$$ Definition of Laplace Transform

$$= \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} e^{-st} f(t)\ \mathrm{d}t$$ Sum of Integrals on Adjacent Intervals for Integrable Functions: Corollary

$$= \sum_{k=0}^{\infty} \int_0^T e^{-s(t+kT)} f(t+kT)\ \mathrm{d}t$$ Change of Limits of Integration

$$= \sum_{k=0}^{\infty} \int_0^T e^{-s(t+kT)} f(t)\ \mathrm{d}t$$ General Periodicity Property

$$= \sum_{k=0}^{\infty} \int_0^T e^{-st-skT} f(t)\ \mathrm{d}t$$ Real Multiplication Distributes over Addition

$$= \sum_{k=0}^{\infty} \int_0^T e^{-st} e^{-skT} f(t)\ \mathrm{d}t$$ Product of Powers

Proof continued

$$= \sum_{k=0}^{\infty} e^{-skT} \int_0^T e^{-st} f(t) \, \mathrm{d}t$$

Primitive of Constant Multiple of Function

$$= \left(\int_0^T e^{-st} f(t) \, \mathrm{d}t \right) \sum_{k=0}^{\infty} e^{-skT}$$

Scaling of Summations

$$= \left(\int_0^T e^{-st} f(t) \, \mathrm{d}t \right) \sum_{k=0}^{\infty} \left(e^{-sT}\right)^k$$

Power of Power

$$= \left(\int_0^T e^{-st} f(t) \, \mathrm{d}t \right) \left(\frac{1}{1 - e^{-sT}} \right)$$

Sum of Infinite Geometric Sequence

$$= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) \, \mathrm{d}t$$

Real Multiplication is Commutative

Example 1 Let $f(x) = \sin ax,\ a > 0$. Then $f(x)$ has period $X = \frac{2\pi}{a}$ and we must have

$$(\mathcal{L}\sin ax)(s) = \frac{\int_0^{\frac{2\pi}{a}} e^{-sx} \sin ax\, dx}{1 - e^{-\frac{2\pi s}{a}}}$$

$$= \frac{\left(-\frac{a}{s^2+a^2} e^{-sx} \cos ax - \frac{s}{s^2+a^2} e^{-sx} \sin ax\right)\Big|_0^{\frac{2\pi}{a}}}{1 - e^{-\frac{2\pi s}{a}}} = \frac{\frac{a}{s^2+a^2}\left(1 - e^{-\frac{2\pi s}{a}}\right)}{1 - e^{-\frac{2\pi s}{a}}}$$

$$= \frac{a}{s^2 + a^2}.$$

Example 2 The *square wave* of period $2X$ is the $2X$-periodic function $f(x)$ whose basic function is

$$f_{2X}(x) = \begin{cases} 1, & 0 \leq x < X; \\ -1, & X \leq x < 2X. \end{cases}$$

Thus $f(x)$ has the value $(-1)^k$ on $kX \leq x < (k+1)X, \ k = 0, 1, 2, \cdots$. From the formula we have

$$(\mathcal{L}f)(s) = \frac{\int_0^X e^{-sx} \cdot 1 \, dx - \int_X^{2X} e^{-sx} \cdot 1 \, dx}{1 - e^{-2sX}} = \frac{1}{s} \frac{(1 - 2e^{-sX} + e^{-2sX})}{1 - e^{-2sX}}$$

$$= \frac{1 - e^{-sX}}{s(1 + e^{-sX})}.$$

Example 3 A periodic function with numerous applications in design of electronic devices is the so-called *saw-tooth function*. The basic function is $f_X(x) = x,\ x \in [0, X)$ and $f(x)$ is then obtained by periodic continuation with period X; on the interval $[kX, (k+1)X)$ we have $f(x) = x - kX$. In this case

$$(\mathcal{L} f_X)(s) = \int_0^X e^{-sx} f(x)\, dx = \int_0^X e^{-sx}\, x\, dx = -\frac{d}{ds} \int_0^X e^{-sx}\, dx$$

$$= -\frac{d}{ds} \frac{1 - e^{-sX}}{s} = \frac{1 - e^{-sX}}{s^2} - \frac{X e^{-sX}}{s}.$$

Then

$$(\mathcal{L} f)(s) = \frac{1}{s^2} - \frac{X e^{-sX}}{s(1 - e^{-sX})}.$$

Example 4 We consider a *periodic pulse*, i.e., an intense signal of short duration emitted at periodic intervals. We again let the period be $X > 0$ and we suppose the basic function, given in terms of an amplitude, A, $A > 0$ and a pulse duration h, $0 < h < X$, is

$$f_X(x) = \begin{cases} A, & 0 \le x < h; \\ 0, & h \le x < X. \end{cases}$$

Then

$$\int_0^X e^{-sx} f_X(x)\, dx = A \int_0^h e^{-sx}\, dx = \frac{A}{s}(1 - e^{-sh}).$$

Taking $f(x)$ to be the X-periodic extension of $f_X(s)$, our formula gives

$$(\mathcal{L}f)(s) = \frac{A(1 - e^{-sh})}{s(1 - e^{-sX})}.$$

Thanks all